



CENTRE FOR ADVANCED STUDIES

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY



2-Days Workshop on

Metal Failures due to Corrosion

Highlights



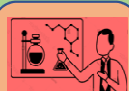
Certificate
for
attendance



Friendly
Discussion



Experts led
Training



Hands on
training

Note: Accommodation facility and other financial expenses will not be provided in any case.

Registration Link:

<https://forms.gle/VVugJwZanzuf2xna7>



Coordinator

Dr. Gopal Ji

Assistant Professor
Centre for Advanced Studies
Dr. A.P.J. Abdul Kalam
Technical University, Lucknow

Eligibility: B. Tech and M.Tech (ME/ MME/ AC/CE), B.Sc/M.Sc chemistry students and all interested persons, **Date :** 26-27 June 2023, **Mode:** Offline/Online, **Number of Seats:** 50

Registration Fee – Free

Limited seats (First Come First Serve)

Last Date of Registration : 15 June 2023

Contents

- ✓ Corrosion Definition and examples, Origin of corrosion, Basics of Corrosion
- ✓ Types of corrosion, detailed discussion about each types Need for corrosion control, Corrosion Control Methods
- ✓ Electrochemical Measurements: Basics.
- ✓ Detailed discussion about EIS, Equivalent electrical Circuit formation and their physical significance.
- ✓ Tafel Polarization curves: Basics and interpretation.
- ✓ Linear Polarization Resistance and OCP .
- ✓ Role of electrochemical analysis in corrosion study.

After Completion The Workshop

- ✓ You will be able to know how to start work in corrosion
- ✓ UG/PG/Ph.D students will be able to understand the basics of corrosion
- ✓ PG and PhD students will be able to write the manuscript on corrosion more effectively and scientifically.
- ✓ Better understanding of electrochemical analysis of corrosion.



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2-Days Workshop on Metal Failures due to Corrosion

Days	Schedule
Day 1 (Forenoon)	Introduction and Basics of Corrosion: Origin of Corrosion, Reasons for Corrosion, Basic Concepts in Corrosion, Electrochemical theory of corrosion, Types of corrosion: galvanic, Crevice, Pitting, SCC, Intergranular, Selective Leaching, Erosion Corrosion, Corrosion fatigue, microbial Corrosion, Pipeline Corrosion , Need of corrosion prevention, Corrosion Control Methods: Material Selection, Design, Coating, Cathodic protection, Inhibitors,
Day 1 (Afternoon)	Electrochemical Corrosion Monitoring: Basics, OCP, Key findings based on OCP, Introduction to Polarization, Types of Polarization, Over potential, Tafel Polarization curves , Exchange and Corrosion Current Density, Mixed potential Theory, Ecorr, Icorr, Tafel Extrapolation, Key findings based on Tafel curves, Solved examples ,
Day 2 (Forenoon)	Electrochemical Corrosion Monitoring : Impedance, Basics, Nyquist and Bode Curves, Curve Fitting, EEC, EEC formulation, Physical significance Interface analysis, Solved examples
Day 2 (Afternoon)	Hands-on Training Sessions/ Discussion: Electrochemical Set up, OCP tests, Tafel polarization curves test , Linear polarization resistance finding, Electrochemical Set up for EIS, Impedance test, Nyquist and Bode plots explanation, Basic Curve fitting



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